

EC Checklist/Rubric:

- Make sure properties that appear in the element don't appear in the custom element before the UP that populates them
- Make that tool outputs to populate components appear before components are called
- Make sure that there are no direct user interaction components
 - This includes components that "look" like they would be interactable, even if they're not actually (e.g. "create", "edit")
- Tools need to have descriptions / definitions
 - Same tool must maintain its same definition if used multiple times in the data
 - **Where do these live in the data?**
- If a user prompt claims to have done something, we need proof that it was actually done
 - E.g. "I've uploaded a PDF"
 - **How do we evidence this? Include an actual PDF?**
- "Expected Components" section of Grading Guidance should be natural language, not a strict list
 - Should also only be present if the components are actually expected in response to that prompt - may have some GG with no expected components
- Any time that some functionality or action is claimed by the model, it must actually provide the tool calls to do so
 - E.g. "Alerts via email are active" but this isn't actually set up and no tool calls are present
 - See if any external actions are taken outside of conversation and if there was, is there a tool call for it
- Avoid placeholder names/emails (e.g. [@example.com](#))
- No empty assistant messages or user messages should appear after tool responses - tool responses must be followed by a summary or quick closing statement from the assistant
- If a tool message is supposed to generate a UI component, it must be followed by that component
- If a link is given, there should be a preceding web search to obtain the correct link
 - If it's something that would have changed the past year or is more obscure, then need the tool call (e.g. wouldn't need it for [Spotify.com](#) but would need for unsubscribe link)
- Components should not be generated when not relevant to the most recent user prompt/response
 - Or when redundant - if assistant response or component contains the info, the other does not need to also
- In GG, expected components need to fit the current prompt, shouldn't have any expected if a clarification is needed or a component isn't actually necessary to be provided
- Tools to extract or format data only should not exist - the model can do this itself
- Components should be display-only - no clickable elements or what looks clickable
- Grading guidance should be tailored toward what is expected from that prompt, not from the whole conversation

- If a clarifying question is expected, the grading guidance should reflect that
 - **What were the specific examples here that looked like they were for the whole conversation?**
- If a response says that it gives X (e.g. a component), it needs to actually give X
 - E.g. “Here is your weekly matrix, load, fairness score” and doesn’t actually contain these
 - If this occurs, then just edit assistant response to not say that it’s included
 - **WHAT IF THE USER PROMPT EXPLICITLY ASKS FOR THIS?**
- If there are expected components given, they must be used
 - Can be given in natural language as “must”, “nice”, etc.

Per sample feedback

- SIMPLE_household_chore_rotation
 - turn 1: GG needs to be tailored to what we expect now, not at the end of the conversation; here we’re expecting clarifying questions from the model and thus this should be mentioned (+ custom elements not needed at this stage) ✓
 - tool descriptions needed ✓
 - “Great \u2014 here are Weeks 2\u20134 with weekly matrices, loads, fairness scores, highlights,...” ⇒ the response doesn’t actually contain the matrix/component. ✓
 - “Here\u2019s your printable one-pager for Weeks 1\u20134, plus .ics calendar files per ...” ⇒ the answer contains the elements that should be in the custom element already; better not to have redundant info. **Only show relevant UI elements** ✓
 - in the last turn: the assistant doesn’t actually provide a PDF - so it shouldn’t state that it does. ics sort of fine as the model gives the text to copy paste and actually tells the user to copy paste, but still not ideal. **Relates to question asked above ^^**
- COMPLEX_freelance_client_invoice_organiser
 - turn 1: GG isn’t tailored to it (as above) ✓
 - expected components
 - should only be mentioned when actually expected;
 - and not always in list format ✓
 - turn 2: parse csv makes little sense as the user shared the data in the UP, not as an attached csv ☐
 - this is a complex sample, but it’s not clear where there are nested components or composition of components; the filter seems stand alone rather than acting on a table/another component. **Relates to question above**
 - the tools calls seem to mostly reformat the data ✓

- tool definitions missing ✓
- avoid [me@example.com](#) ✓
- COMPLEX_daily_focus_and_time_blocking_assistant
 - turn 1: again, GG is not adjusted
 - turn 2: there are expected components, yet none is used. This is also why we want the expected_components to be in natural language so it can be mentioned which are **must-haves, nice-to-haves**, when you may not expect one and why, etc. **Would be curious about what this looks like exactly, esp related to 'when you may not expect one and why'?**
 - last turn: not clear where the reminder is set - the model said it did, but there was no associated tool call **All actions must have associated tool call? Or which types of actions?**
 - custom components seem overly specific **? examples of overly specific vs what they want? A schedule visualization seems pretty normal to me**
 - no tool descriptions;
 - no nested components
- COMPLEX_content_calendar_for_newsletter ✓
 - turn 1: GG should say that we're expecting a clarifying question here.
 - turn 1: the model is just hallucinating info given that the user didn't provide any yet
 - no nested components
 - no tool descriptions
- Do not want properties appearing in custom element before UP that allows to populate them
 - Hard to make LLM not hallucinate content that gets added to components
 - Try to make it impossible for it to do this
 - Add more prompting to ensure that the LLM only uses context provided by tool call
- Custom components should not contain elements that allow for direct user interaction
 - Update component generation part
 - Just display information, not collect information
- Need to watch out for logical order
 - Tool outputs needed to populate components should appear before the components are called by the model
 - Reordering sequence should take care of this
- Conversation is too big for evaluator to check all at once
 - Have it run an eval for each turn (each tool/message pair)
- Update prompt for LLM in GG generation - switch order so it appears before assistant response after UP

Tools

- Tools are not very realistic and are missing clear definitions:
 - We need in each sample the tool definitions
 - Some tools are used differently within the same conversation - this would be easier to judge if tool definitions were provided.
 - There may have been a misunderstanding following our previous feedback. We don't expect tools to format the data to what will be needed to populate a custom component. We merely meant that any info used to populate a custom component should be taken or inferred from somewhere in the earlier conversation context (and not hallucinated) => no need for tools reformatting existing data or parsing CSVs, models can do that on their own
 - Make sure sequential tool calls are defined sequentially (i.e. in distinct assistant messages; all tool calls that appear in the same assistant message are expected to be run in parallel and so one cannot be based on the tool output of the other)
 - is the tool_executed = true actually true?
- Need to provide tool definitions for each sample, definitions need to remain uniform so that same tool is used in same way in the conversation
 - SYNTHETIC
- Tools to format data for component are not needed, just need to provide information that is then used for component
 - SYNTHETIC + MANUAL
- Sequential tool calls should appear in distinct assistant messages, if in same message then they are assumed to be running in parallel
 - SYNTHETIC + MANUAL

Conversation flow:

- in some samples (e.g. SIMPLE_subscription_cleanup) we go from a tool output to a user message; this isn't possible. There must be an assistant message with text before a new UP can be issued.
- Need to make sure that flow is always logical: UP -> TC -> TO -> AR, assuming Tool is used in that turn
 - ADD CHECK

Components:

- In the complex conversations, the components do not have nested components.
 - CHECK THIS, I THINK THEY ALL DID
- Could be made a bit more generalizable. You can take inspiration from <https://widgets.chatkit.studio/gallery> for the level of granularity of components we expect. Note that this is just for inspiration! The components shouldn't be reused as is. We want diversity and don't want to overfit on a short list of custom components either.
- Reminder: Avoid buttons like "create", "edit" which expect an immediate response/action from the assistant (as per previous feedback).

- Not only should elements not be clickable, we should avoid any buttons that appear that they would be clickable
 - SYNTHETIC + MANUAL
- Components should be more generalizable
 - ASK QUESTION ON THIS - WHAT WERE SPECIFIC EXAMPLES OF THINGS NOT BEING GENERALIZABLE/WHAT IS THE DEFINITION OF GENERALIZABLE

Grading guidances:

- Grading Guidance should be adapted to each user message including the “expected_components” section. For instance most conversations start with a UP that requires a clarifying question from the model so the GG should mention that and the expected_components should likely stay empty for that turn.
- expected_components, as per the guidelines, should not be a strict list but mention in natural language which components might be expected, making sure to take the exact name of the component.
- For UP that don't require components, expected components should be empty
 - LLM tends to always generate expected components
 - MANUAL
- Don't give list of expected components, describe in natural language (but use exact component name)
 - SYNTHETIC

User/model interactions:

- Some actions from the model are not reasonable (sending - not just drafting - emails, etc.)
- Also, some actions are claimed but not based on any tool call (e.g. google_calendar_schedule_event to set a reminder event)
- sometimes the user or assistant say a pdf / file was attached or created when this isn't the case.
- Please avoid using ...@example.com and other LLM-generated entities.
- Sample would need to be quite enterprise focused (ok here - but mentioning it as not in previous guidelines).
- Actions need to be reasonable
 - SOMEWHAT SYNTHETIC + MORE MANUAL
 - QUESTION: WHAT DEFINES A REASONABLE ACTION?
 - Make sure starting scenarios don't lead to improper interactions (e.g. requiring uploading PDF) - can't have anything that would require attaching file
 - MOSTLY SYNTHETIC + MANUAL
- Placeholder names/emails shouldn't be used
 - SYNTHETIC + MANUAL

- CHECK - EVERY ACTION TAKEN BY THE MODEL HAS A DEFINED ASSOCIATED TOOL CALL

PIPELINE FLOW: Generate offline -> LLM checks offline -> upload to platform -> edits -> evals on platform -> export from platform as packaged data -> run LLM checks again

- Put stuff in platform that violates LLM checks or not? No, just use stuff that doesn't violate any of the checks

Thanks for this detailed feedback. Please see our responses/questions to some specific feedback points below: